

TDDE70 Lab2 – Diffusion Models

Preparatory exercises

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1 Background

In the lab, you will create your own implementation of a Denoising Diffusion Probabilistic Model (DDPM) and see how it can act as a powerful generative model. In the preparation exercises below, you will derive some of the expressions that have been presented in the lectures and that you will use in your implementation.

2 Questions

Model definition

In DDPM, we assume a *noising process* that takes data $\mathbf{x}_0$ and transforms it into noise $\mathbf{x}_T$ via a Markovian iterative process defined by

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) \stackrel{\text{def.}}{=} \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}), \quad (1)$$

where $\{\beta_t\}_{t=1}^T$ is a user-defined *variance-schedule*.

Question 1:

- a) Which bounds do we need to put on the $\{\beta_t\}_{t=1}^T$ variables?
- b) Write down a mathematical expression for $\{\beta_t\}_{t=1}^T$, assuming a linear schedule that starts from $\beta_1 \stackrel{\text{def.}}{=} \beta_{\min} = 0.0001$ and ends at $\beta_T \stackrel{\text{def.}}{=} \beta_{\max} = 0.02$ with $T = 1000$ steps. How do you get the schedule in python? (Hint: look up `torch.linspace`)

Model training

During training, we have to sample a noisy image from a random step t , but doing this according to the forward process in Eq. (1) step by step¹ would take too much time. An equivalent way is to sample each noisy image directly from its conditional distribution given the initial state $\mathbf{x}_0$:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (2)$$

Question 2: Derive the expression of $\bar{\alpha}_t$ in terms of $\{\beta_t\}$.

Hint: Start by deriving $q(\mathbf{x}_2 | \mathbf{x}_1)$ as a linear transformation of the Gaussian variable $\mathbf{x}_1 | \mathbf{x}_0$, and then repeat.

¹That is, sampling $\mathbf{x}_1 \sim q(\mathbf{x}_1 | \mathbf{x}_0)$, then $\mathbf{x}_2 \sim q(\mathbf{x}_2 | \mathbf{x}_1)$, ..., then $\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_{t-1})$.

Generative procedure, posterior

As mentioned in the lectures, to construct a *generative* procedure, we try to invert the noising process, i.e., estimating

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \int q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0. \quad (3)$$

The first factor in the integral, $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ is known on closed form as

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}). \quad (4)$$

where the full expression has been given at the lecture.

Question 3: Derive the expression of $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0)$ and $\tilde{\beta}_t$.

Hint: Take a look at the example "Linear Gaussian measurement update" in the "TDDE70 - basic-probability.pdf" document.

Question 4: The expression that we have derived for $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ is defined for any $t \geq 2$. However, for convenience, we can extend the definition to apply also at $t = 1$ (i.e., since the distribution is over $\mathbf{x}_{t-1}$ we get a distribution over the clean image $\mathbf{x}_0$ when $t = 1$). How should we define $\bar{\alpha}_0$ for the expression derived above to apply also for $t = 1$?

Generative procedure, transition

In the reverse process, at each step, we first predict some noise $\hat{\epsilon} = \epsilon_\theta(\mathbf{x}_t, t)$ by a neural network, which we can use to "reconstruct" $\mathbf{x}_0$ as $\hat{\mathbf{x}}_0$. We can now replace $q(\mathbf{x}_0|\mathbf{x}_t)$ in Eq. (3) with a point-estimate $\delta_{\hat{\mathbf{x}}_0}(\mathbf{x}_0)$ as an approximation. Doing this, we obtain our (reverse-time) generative process

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \stackrel{\text{def.}}{=} \int q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)\delta_{\hat{\mathbf{x}}_0}(\mathbf{x}_0)d\mathbf{x}_0. \quad (5)$$

Question 5: Derive the expression of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ as a function of $\epsilon_\theta(\mathbf{x}_t, t)$, and verify that this corresponds to the generation algorithm in the lectures.

Hint: : You can first derive an estimated $\hat{\mathbf{x}}_0$ from the predicted $\epsilon_\theta(\mathbf{x}_t, t)$ (see Eq. (2)). And then compute a posterior of $\mathbf{x}_{t-1}$ conditioned on $\mathbf{x}_t$ and $\hat{\mathbf{x}}_0$ (as you derived in Question 3).